The Scientific Revolution

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The Scientific Revolution was a significant movement in scientific ideas that happened in the sixteenth and seventeenth centuries. A new concept of nature emerged during the Scientific Revolution, replacing the Greek approach that had dominated science for nearly 2,000 years. Science developed as a distinct topic distinct from both philosophy and technology, and it began to be thought to serve utilitarian ends.

Sir Isaac Newton's work symbolizes the peak of the Scientific Revolution towards the end of the 17th century. His massive Philosophiae Naturalis Principia Mathematica (1687; Mathematical Principles of Natural Philosophy) addressed the fundamental mechanical and cosmological issues brought by the Scientific Revolution. It established the issues and techniques that dominated much of astronomy and physics for well over a century, provided a physical basis for Kepler's laws, unified celestial, and terrestrial physics under one set of principles, and established the issues and methods that dominated much of astronomy and physics for well over a century. Through the concept of force, Newton was able to combine two basic components of the Scientific Revolution: mechanical philosophy and nature mathematization. To apply his laws to astronomy, Newton had to go above Descartes' limitations. He hypothesized a gravitational force acting between any two objects in the universe, but he couldn't explain how it could be conveyed. Using his equations of motion and a gravitational force proportionate to the inverse square of the distance between the centers of two bodies, Newton was able to calculate Kepler's laws of planetary motion. Galileo's law of free fall agrees with Newton's laws. The same force that causes objects to fall near the Earth's surface is also responsible for keeping the Moon and planets in their orbits (Scientific Revolution | Definition, History, Scientists, Inventions, & Facts, 2019).

The growing flow of information that resulted from the Scientific Revolution put immense strain on conventional institutions and conventions. It was no longer sufficient to publish scientific results in a pricey book that few could afford; information needed to be communicated widely and fast. Natural philosophers wanted to be certain of their data, therefore they sought independent and critical validation of their findings. To attain these objectives, new methods were devised.

References

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